

Connor Wilson

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Professional Experience

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| Doctoral Researcher
Northeastern University (Advisor: Dr. Cody Dunne) | September 2022 – Present
Boston, MA |
| <ul style="list-style-type: none">• Conducted and published novel research on fast, scalable techniques for layered network visualization, as well as computational benchmarking techniques and datasets for network layout algorithms.• Created and maintained python library that uses optimization models to generate graph visualizations.• Worked on research both independently and as part of multidisciplinary teams to publish qualitative and quantitative study results, research tools, and datasets.• Taught CS7250 Advanced Information Visualization, DS4200 Data Visualization (3×) as a teaching assistant. | |
| Research Engineer
Kostas Research Institute | June 2025 – September 2025
Boston, MA |
| <ul style="list-style-type: none">• Created interactive, fully custom data visualizations and front-end design for global supply chain mapping application using d3.js and Svelte.js.• Optimized database accesses to significantly improve application load time.• Implemented network algorithms and PostgreSQL database schema for 1TB global road network dataset. | |
| Research & Teaching Assistant
University at Buffalo (Advisor: Dr. Erdem Sariyüce) | September 2020 – May 2022
Buffalo, NY |
| <ul style="list-style-type: none">• Implemented and improved algorithms for analyzing network substructures in a financial transaction dataset.• Taught CSE331 Algorithms & Complexity (4×). | |
| Quality Engineering Intern
American Packaging Corporation | June 2018 – August 2020
Rochester, NY |
| <ul style="list-style-type: none">• Conducted Gage Repeatability & Reproducibility studies and ANOVA for quality assurance procedures. | |

Education

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| Ph.D. in Computer Science
Northeastern University, Boston, MA | Expected June 2027 |
| M.S. in Computer Science
Northeastern University, Boston, MA | June 2024 |
| B.S. in Mathematics & B.S. in Computer Science
University at Buffalo, Buffalo, NY | June 2022 |

Publications

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5. **Fast and Readable Layered Network Visualizations Using Large Neighborhood Search**
Wilson, C., Crnovrsanin T., Puerta E., and Dunne C. *IEEE Transactions on Visualization and Computer Graphics*. 2025. doi.org/10.1109/TVCG.2025.3537898.
 4. **Evaluating and extending speedup techniques for optimal crossing minimization in layered graph drawings**
Wilson, C., Puerta E., Di Bartolomeo S., Crnovrsanin T., and Dunne C. *IEEE Transactions on Visualization and Computer Graphics—VIS/TVCG*. 2024. [10.1109/TVCG.2024.3456349](https://doi.org/10.1109/TVCG.2024.3456349).
 3. **Evaluating Graph Layout Algorithms: A Systematic Review of Methods and Best Practices**
Di Bartolomeo S., Crnovrsanin T., Saffo D., Puerta E., *Wilson, C.*, and Dunne C. *Computer Graphics Forum—EuroVis/CGF*. 2023. doi.org/10.1111/cgf.15073.
 2. **A collection of benchmark datasets for evaluating graph layout algorithms**
Di Bartolomeo S., Puerta E., *Wilson, C.*, Crnovrsanin T., and Dunne C. *Poster at Graph Drawing and Network Visualization—GD Posters*. 2023. **Best Poster Honorable Mention*

1. Indy Survey Tool: A framework to unearth correlations in survey data

Crnovrsanin T., Di Bartolomeo S., **Wilson, C.**, and Dunne C. *Proc. IEEE Visualization Conference—VIS*. 2023. doi.org/10.1109/VIS54172.2023.00038.

Under Submission

- **CINDER: An expressive framework for optimal layered graph layouts**
Wilson, C., Puerta E., and Dunne C. *Under submission to IEEE Transactions on Visualization and Computer Graphics*.
- **GLaDOS: Graph Layout algorithm Datasets for Open Science**
Di Bartolomeo S., **Wilson, C.**, Puerta E., Crnovrsanin T., Frings A., and Dunne C. *Under submission to the Journal of Visualization and Interaction*.

Service & Volunteer Work

Conference Student Volunteer	2024 – 2025
IEEE VIS 2024, Tampa, FL; IEEE VIS 2025, Vienna, Austria	
Lab Operations Manager	2022 – 2025
Data Visualization Lab, Northeastern University, Boston, MA	
Conference/Journal Reviewer	2024
IEEE PacificVis; SoftwareX	
PhD Program Application Committee Member	2023
Khoury College of Computer Science, Northeastern University, Boston, MA	

Technical Skills & Interests

Programming Languages: Python | JavaScript | C | HTML | CSS

Technologies and Database Systems: Gurobi | React | Svelte | d3.js | Jupyter | SQL | MongoDB | PgAdmin

Interests: Running | Chess Puzzles | Electronic Music | Specialty Coffee | Buffalo Bills Fan | Mentoring